INCH-POUND

MIL-R-122A
AMENDMENT 2
8 March 1993
SUPERSEDING
AMENDMENT 1
26 May 1992

MILITARY SPECIFICATION

RESISTORS, FIXED, PRECISION, ESTABLISHED RELIABILITY,

GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-122A, dated 19 February 1990 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.1, delete and substitute:

"1.2.1 Part or Identifying Number (PIN). The resistors specified herein shall be identified by a PIN which shall consist of the basic military specification sheet and a coded number. The number will be coded to provide information concerning the reactance characteristic, resistance tolerance and failure rate, resistance value, decimal point location, and temperature characteristic. The PIN shall be in the following form with the coded number derived as indicated."

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2.1.1, following MIL-STD-1285, add the following:

"MIL-STD-45662 - Calibration Systems Requirements."

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3.2, delete and substitute:

"3.2 Qualification. The resistors furnished under this specification shall be products which are qualified for listing on the applicable qualified product list (QPL) at the time set for opening bids (see 4.4 and 6.3). In addition, the manufacturer shall obtain certification from the qualifying activity that the reliability assurance requirements of 4.1.2 have been met and are being maintained. Authorized distributors that are approved to MIL-STD-790 distributor requirements by the QPL manufacturer are listed in the QPL."

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3.5.3, add the following new paragraphs:

"3.5.3.1 Solder dip (retinning) leads. The manufacturer (or his authorized category B or C distributor) may solder dip/retin the leads of product supplied to this specification provided the solder dip process has been approved by the qualifying activity.

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"3.5.3.1.1 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options:

a. When the original qualified lead finish was hot solder dip lead finish 52 of MIL-STD-1276. (NOTE: The 200-microinch maximum thickness is not applicable.) The manufacturer shall use the same solder dip process for retinning as was used in the original manufacture of the product.

b. When the original qualified lead finish was not hot solder dip lead finish 52 of MIL-STD-1276 as prescribed in 3.5.3.1.1a., approval for the process to be used for solder dip shall be based on the following test procedures:

(1) Thirty samples of any resistance value for each style and lead finish are subjected to the manufacturer's solder dip process. Following the solder dip process, the resistors shall be subjected to the direct current resistance test and other group A electrical tests. No defects are allowed.) (NOTE: If hermetic seal testing is required in group A, these tests shall also be performed.) No defects allowed

(2) Ten of the 30 samples are then subjected to the solderability test. No defects are allowed.

(3) The remaining 20 samples are subjected to the resistance to solder heat test followed by the moisture resistance test (or hermetic seal test if the device is hermetically sealed). No defects are allowed.

"3.5.3.1.2 Solder dip/retinning options. The manufacturer (or the authorized category B or C distributor) may solder dip/retin as follows:

a. After the 100-percent group A screening tests and following the solder dip/retinning process, the electrical measurements required in group A, subgroup I, 100-percent screening tests shall be repeated on 100 percent of the lot. (NOTE: The manufacturer may solder dip/retin prior to the 100 percent electrical measurements of the group A, subgroup I tests.) Lot rejection criteria shall be the same as for subgroup I. (NOTE: If hermetic seal is required in group A, subgroup I, this test shall be performed after the solder dip/retinning process or the tests will have to be repeated.)

b. As a corrective action, if the lot fails the group A solderability test.

c. After the group A inspection has been completed and following the solder dip/retinning process, the electrical measurements required in group A, subgroup I, 100 percent screening test shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be the same as for subgroup I tests. (Note: If hermetic seal testing is required in the group A, subgroup I tests, these tests shall be repeated.) Following these tests, the manufacturer shall submit the lot to the group A solderability test as specified in 4.7.7.

* "3.5.3.1.2.1 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat (see 6.3.1). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent."

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4.4.5, last sentence, delete in its entirety.

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TABLE VIII, subgroup II, Mechanical (ppm-3), requirement paragraph column: Delete "3.4" and substitute "3.5".

TABLE VIII, add the following new subgroup:

"

Inspection	Requirement paragraph	Test method paragraph	Number of samples
Subgroup IV			
Solderability	3.13	4.7.7	13

"

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4.6.1.2.4, delete and substitute:

"4.6.1.2.4 Subgroup III tests. A sample of parts shall be randomly selected in accordance with table IX. If one or more defects are found, the lot shall be screened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table IX. If one or more defects are found in the second sample, the lot shall be rejected and the lot shall not be supplied to this specification."

Add the following new paragraphs:

"4.6.1.2.5 Subgroup IV (solderability).

"4.6.1.2.5.1 Sampling plan. Thirteen samples shall be selected randomly from each inspection lot and subjected to the subgroup 4 solderability test. The manufacturer may use electrical rejects from subgroup I screening tests for all or part of the samples to be used for solderability testing. If there are one or more defects, the lot shall be considered to have failed.

"4.5.1.2.5.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot:

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.7.7 production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure in 4.5.1.2.5.2b.

b. The manufacturer submits the failed lot to a 100-percent solder dip using an approved solder dip process in accordance with 3.5.3.1. Following the solder dip, the electrical measurements required in group A, subgroup 1 tests shall be repeated on 100 percent of the lot. The percent defective allowable for the electrical measurements shall be the same as for the subgroup I tests. Thirteen additional samples shall then be sujected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot shall be considered rejected and shall not be supplied to this specification.

"4.6.1.2.5.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on this contract."

Table IX, delete and substitute:

"TABLE IX. Sample sizes for subgroup 2 and 3.

Lot size	Subgroup 2 samples size	Subgroup 3 sample size
1 to 8	100 percent	100 percent
9 to 90	100 percent	13
91 to 150	125	13
151 to 280	125	20
281 to 500	125	29
501 to 1,200	125	34
1,201 to 3,200	125	42
3,201 to 10,000	192	50
10,001 to 35,000	294	60
35,001 to 150,000	294	74
150,001 to 500,000	345	90
500,001 to and over	435	102

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4.6.1.3.1.1, delete and substitute:

"4.6.1.3.1.1 Subgroup I. A sample of parts shall be randomly selected in accordance with table Xa. If one or more defects are found, the lot shall be screened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table Xa. If one or more defects are found in the second sample, the lot shall be rejected and the lot shall not be supplied to this specification."

4.6.1.3.1.2, delete and substitute:

"4.6.1.3.1.2 Subgroup II. A sample part shall be randomly selected in accordance with table Xa. If one or more defects are found, the lot shall be screened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table Xa. If one or more defects are found in the second sample, the lot shall be rejected and the lot shall not be supplied to this specificaiton."

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Table X, delete and substitute:

"TABLE X. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of samples
Subgroup I			
Resistance temperature characteristic	3.15	4.7.9	see 4.6.1.3.1.1
Dielectric withstanding voltage (atmospheric)	3.18	4.7.12	
Insulation resistance	3.19	4.7.13	
Subgroup II			
Visual and mechanical inspection (when applicable) 1/	3.1, 3.31, and 3.32	4.7.1	see 4.6.1.3.1.2

1/ Only applicable when marking inspection is not performed in group A inpspection. No defects shall be permitted. Marking shall remain legible at end of all tests."

Add the following new table:

"TABLE Xa. Group B sampling plan.

Lot size		Sample size
1 to	50	5
51 to	90	7
91 to	150	11
151 to	280	13
281 to	500	16
501 to	1,200	19
1,201 to	3,200	23
3,201 to	10,000	29
10,001 to	35,000	35
35,001 to	150,000	40
150,001 to	500,000	40
500,001 and over		40

"

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4.7.5, delete and substitute:

"4.7.5 DC resistance (see 3.11). Resistors shall be tested in accordance with method 303 of MIL-STD-202. The following details and exception shall apply:

a. Measuring apparatus: Different types of measuring test equipment (multimeter, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same style, model, or if it can be shown that the performance of the equipment is equivalent. All test equipment shall be calibrated in accordance with MIL-STD-45662.

b. Limit of error of measuring apparatus shall not exceed one-fourth of the resistance tolerance or the resistance change limit for which the measurement is being made. Manufacturers, at their option, may use the apparatus of less accuracy, provided limits are reduced to fully compensate for accuracy deviation.

c. Test voltage for bridges: Measurements of resistance shall be made by using the test voltages specified in table XII. The test voltage chosen, whether maximum or a lower voltage which would still provide the sensitivity required, shall be applied across the terminals of the resistor. This same voltage shall be used whenever a subsequent resistance measurement is made.

d. Measurement energy for electronic test equipment: The measurement energy applied to the unit under test shall not exceed 10 percent of the +25°C rated wattage times 1 second.

e. Resistance measurement point: See 3.1.

f. Temperature: The dc resistance test specified in group I of table VII shall be performed at +25 ±2°C. For all other tests, unless otherwise specified herein, the temperature at which subsequent and final resistance measurements are made in each test shall be made within ±2°C of the temperature at which the initial resistance measurement was made."

TABLE XIII, delete and substitute:

"TABLE XIII. Resistance temperature characteristic.

Sequence 1/	Temperature
	Qualification and group inspection
	°C
1	Room temperature 2/
2	-15 ±3
3	-55 ±3
4	Room temperature 2/
5	+65 ±3
6	+125 ±3
7	Maximum temperature (see 3.1)

1/ NOTE: At the option of the manufacturer, the reverse sequence of table XIII may be as follows:

1. Room temperature.
2. Series of high temperatures.
3. Room temperature.
4. Series of low temperatures.

2/ Reference temperature for each of the succeeding temperatures."

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6.3, add the following new paragraph:

* 6.3.1 Tin plated finishes. Tin plating is prohibited (see 3.5.3.1.2.1), since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter, refer to ASTM B545."

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CONCLUDING MATERIAL

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Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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